

REG NO: 22TD0053 NAME: MOHAMED B SIRAJUDEEN
YEAR/SEM/SEC: Y3/S6/B

AIM:

SOURCE CODE:

```
GLOBAL __main
        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX     LR
        ENTRY

;Main loop begins

__main
        LDR R0,=0X00011111 ;load first input
        LDR R1,=0X54711212 ;load second input
        ADD r2,r0,r1

stop    B stop

        end
```

RESULT:

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        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY

;Main loop begins

__main
        LDR R0,=0X00011111 ;load first input
        LDR R1,=0X54711212 ;load second input
        SUB r2,r0,r1

stop      B stop

        end
```

RESULT:

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        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY

;Main loop begins

__main
        LDR R0,=0X00011111 ;load first input
        LDR R1,=0X54711212 ;load second input
        MUL r2,r0,r1

stop      B stop

        end
```

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        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY

;Main loop begins

__main
        LDR  R0,=0x00000006 ;load first input
        LDR  R1,=0x00000004 ;load second input
        ADD  r2,R0,R1
        SUBS R3,R0,R1
        MUL  R4,R0,R1

stop      B stop

        end
```

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GLOBAL __main
        AREA main, CODE, READONLY
        EXPORT __main
        EXPORT __use_two_region_memory
__use_two_region_memory EQU 0
        BX    LR
        ENTRY

;Addresses of Registers
IO0DIR      EQU      0xE0028008
IO1DIR      EQU      0xE0028018
IO0PIN      EQU      0xE0028000
IO1PIN      EQU      0xE0028010
IO0SET      EQU      0xE0028004
IO0CLR      EQU      0xE002800C
IO1CLR      EQU      0xE002801C
IO1SET      EQU      0xE0028014
PINSEL0     EQU      0xE002C000
PINSEL1     EQU      0xE002C004
PINSEL2     EQU      0xE002C014
;

__main

        ldr    r1,=IO1DIR
        ldr    r2,=0x00FF0000      ;initialize input & output
        str    r2, [r1]

loop
        ldr    r1,=IO1PIN
        LDR    r2, [r1]
        LSR    R2,#8
```

```
AND    R2,#0x00ff0000
str     r2,[r1]
B       loop
END
```

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        AREA main, CODE, READONLY
        EXPORT __main
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__use_two_region_memory EQU 0
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        ENTRY

;Addresses of Registers
IO0DIR      EQU      0xE0028008
IO1DIR      EQU      0xE0028018
IO0PIN      EQU      0xE0028000
IO1PIN      EQU      0xE0028010
IO0SET      EQU      0xE0028004
IO0CLR      EQU      0xE002800C
IO1CLR      EQU      0xE002801C
IO1SET      EQU      0xE0028014
PINSEL0     EQU      0xE002C000
PINSEL1     EQU      0xE002C004
PINSEL2     EQU      0xE002C014
LEDDELAY    EQU      800000
;

__main
        ldr    r1,=IO0DIR                ;IO1DIR OUTPUT
CONFIG
        ldr    r2,=0x00FF0000
        str    r2, [r1]

loop
        ldr    r1,=IO0PIN                ;IO1PIN
```

```

ldr    r2,=0x00FF0000
str    r2, [r1]
BL     __delay

ldr    r1,=IO0PIN
ldr    r2,=0x00000000
str    r2, [r1]
BL     __delay

B      loop

```

```

;SOLF DELAY
;Delay loop
;Constant calculations
;1/30Mhz=1/30us=0.03334us
;1us requires 30 counts ie 30x0.0334=1us
;So 1ms requires 30x1000=30000 counts but the SUBS and BNE combination requires 2cy
;So the constant final =30000/2=15000;
__delay

```

```

                                ldr r0,=0x20000
doloop
                                subs r0,#1
                                bne doloop
                                bx lr
                                END

```

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        EXPORT __main
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__use_two_region_memory EQU 0
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        ENTRY
```

;Addresses of Registers

```
IO0DIR      EQU      0xE0028008
IO1DIR      EQU      0xE0028018
IO0PIN      EQU      0xE0028000
IO1PIN      EQU      0xE0028010
IO0SET      EQU      0xE0028004
IO0CLR      EQU      0xE002800C
IO1CLR      EQU      0xE002801C
IO1SET      EQU      0xE0028014
PINSEL0     EQU      0xE002C000
PINSEL1     EQU      0xE002C004
PINSEL2     EQU      0xE002C014
```

;

__main

```
ldr    r1,=IO1DIR      ;IO1DIR
ldr    r2,=0x00FF0000
str    r2, [r1]
```

loop

```
ldr    r1,=IO1PIN      ;IO1PIN
ldr    r2,=0x00FF0000
```

```

str    r2, [r1]
BL     __delay

ldr    r1,=IO1PIN
ldr    r2,=0x00000000
str    r2, [r1]
BL     __delay

B      loop

```

;Delay loop
 ;Constant calculations
 ; $1/30\text{Mhz}=1/30\mu\text{s}=0.03334\mu\text{s}$
 ;1us requires 30 counts ie $30 \times 0.0334=1\mu\text{s}$
 ;So 1ms requires $30 \times 1000=30000$ counts but the SUBS and BNE combination requires 2cy
 ;So the constant final $=30000/2=15000$;

```

__delay

STMDB   sp!, {r14}
LDR     r12, =3000000

LOOP

SUBS    r12, r12, #1
BNE     LOOP
LDMIA   sp!, {r14}
BX      r14

END

```

RESULT: